Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/15/2014
PI Name:	Rubin, David M.S.		
Project Title:	Exploration Medical System Demonstration		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2011	**End Date:**	03/27/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/27/2015 (from 9/30/2018) per PI (Ed., 6/30/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	The Exploration Medical System Demonstration (EMSD) project will act as a test bed on the International Space Station (ISS) to demonstrate to crew and ground personnel that an end-to-end medical system can assist clinician and non-clinician crew members in optimizing medical care delivery and data management during a mission. This system can be designed to assist crewmembers in nominal, non-emergent situations and in emergent situations when they may be suffering from performance decrements due to environmental, physiological or other factors. Any exploration class mission will have medical care challenges, such as resource and resupply constraints limiting the extent to which medical conditions can be treated, inability for crewmembers to evacuate to Earth during many mission phases, and rendering of medical care by a non-clinician. The system demonstrates the integration of medical technologies and medical informatics tools for managing evidence and decision making.		

Task Description:	Development of the EMSD addresses the following risk as stated within the Critical Path Roadmap (CPR; also known as the NASA Bioastronautics Roadmap (NASA/SP-2004-6113)): Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities: Given that medical capabilities will be limited during human exploration missions, there is a possibility that in-flight medical events will lead to unacceptable health and mission outcomes. The objectives of the EMSD are to: a. Reduce time required of on-orbit crew and ground personnel to access, transfer, and manipulate medical data. b. Demonstrate the on-orbit crew's ability to access medical information via an intuitive and crew-friendly solution to aid in the treatment of a medical condition. c. Develop a common data management framework and architecture that can be ubiquitously used to automate repetitive data collection, management, and communications tasks for all activities pertaining to crew health and life sciences. d. Ensure crew access to medical data during periods of restricted ground communication. e. Develop a common data management framework and architecture that allows for scalability, extensibility, and interoperability of data sources and data users. f. Lower total cost of ownership for development and sustainment of peripheral hardware and software that use EMSD for data management. g. Provide a better standard of healthcare for crewmembers by reducing the time required by crew and ground personnel to provide medical treatment and the number of crew errors experienced during treatment. The functionality of the EMSD will be verified through a ground demonstration and a flight demonstration. This project began with market surveys of available Commercial-off-the-Shelf (COTS) solutions for each of the EMSD components, followed by a Request for Proposal process and procurement via the Bioastronautics Contract. On-orbit performance of this demonstration can be accomplished by a single crewmember. Procedures performed by the crewmember will be designed for efficiency by exercising multiple system components concurrently or serially as appropriate. The duration of the demonstration will be dictated by the volume and sources of data necessary to fully exercise the medical capability objectives. Ground performance of this demonstration can be accomplished by a single test executor. Procedures in the ground demonstration will be designed to exercise the system capabilities more thoroughly than during the on-orbit demonstration. The availability of more varied medical condition simulations, more extensive supply of power, fewer time and space limitations, and enhanced system characterization capabilities will allow the ground demonstration to expand the on-orbit objectives by assessing system effectiveness and performance.
Rationale for HRP Directed Research:	This task is the integration of several other tasks within ExMC's integrated research plan. These sub-tasks have separate task synopses (Middleware for Medical System, Dental for Medical System, Biosensors for Medical System, Electronic Medical Records, Assisted Medical Procedures, and Medical Consumables Tracking). The study team is uniquely positioned to integrate these components and prepare the system for ground demonstration and flight demonstration on ISS.
Research Impact/Earth Benefits:	The Exploration Medical System is being designed specifically for a Near Earth Asteroid mission and, as such, has no direct impact or benefits to life on Earth. However, development of the OpenEMR software, required for integration into the Exploration Medical System, can be made available to the OpenEMR open source community for potential inclusion into future releases of the EMR software. In addition, the Exploration Medical System has potential training and simulation opportunities for training of future astronauts and ground personnel. In addition, the Exploration Medical System can serve as software infrastructure for future development and integration of medical devices and software.
Task Progress:	The following accomplishments have been made during the current reporting period of 6/30/13-7/1/14: -The EMSD team dispositioned and implemented all RIDs from Preliminary Design Review process. EMSD project documents were updated and an Authority to Proceed (ATP) was received from the Science Management Office (SMO). -The EMSD team participated in the ExMC System Requirements Review (Phase 3) by submitted updates to the ExMC Operational Concepts (OpsCon) document and Functional Requirements Document (FRD), including a new section for the image and video capture system of EMSD. The EMSD team also participated in the RID review and disposition process for these documents. -The EMSD team has procedure and installed all necessary software on the test computing platform. This includes installation of the EMSD components (assisted medical procedures, electronic medical record, and middleware) and peripheral software (VSee, Cardiax, Windows Server, etc.). The team is currently developing the EMSD component software, as needed, to support full integration of the system. -The EMSD team conducted the Detailed Design Phase of the project. As part of this phase, the EMSD team conducted the Critical Design Review Kick Off and RID Board meetings. All RIDs were dispositioned and implemented and all project documents were updated accordingly. -The EMSD team developed functional and system requirements for the image and video capture system. The team also procured dental cameras as part of this system. -The EMSD team developed an EMSD Validation and Verification Document. -The EMSD team began the Production, Test, and Certification Phase. -The EMSD team integrated the NSBRI SpaceMED system into the EMSD system.
Bibliography Type:	Description: (Last Updated: 04/16/2014)

Abstracts for Journals and Proceedings	Rubin DA, Watkins SD. "Exploration Medical System Demonstration Project." Presented at the 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3047.pdf , Feb-2014
Abstracts for Journals and Proceedings	Shah RV, Rubin DA, Kerstman EL. "NASA Exploration Medical Capability Exploration Medical System Demonstration Project." Presented at the 85th Annual Scientific Meeting, Aerospace Medical Association, San Diego, CA, May 10-15, 2014. Aviation, Space, and Environmental Medicine. 2014 Mar;85(3):327. http://www.ingentaconnect.com/content/asma/asem/2014/00000085/00000003 , Mar-2014